Fine-grained Claim-level RAG Benchmark for Law

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Abstract

The rapid progress of large language models (LLMs) is shifting semantic search toward a question-answering paradigm, where users ask questions and LLMs generate responses. In high-stake domains such as law, retrieval-augmented generation (RAG) is commonly used to mitigate *hallucinations* in generated responses. Nonetheless, prior work shows that RAG systems, whether general-purpose or legal-specific, still hallucinate at varying rates, making fine-grained evaluation essential. Despite the need, existing evaluation frameworks for legal RAG systems lack the granularity required to provide detailed analysis of retrieval and generation performance separately. Moreover, current benchmarks are largely English-only and centered on legal expert queries, overlooking non-expert needs. We introduce ClaimRAG-LAW, a comprehensive dataset for legal RAG that supports French and English, targets both experts and non-experts, and includes diverse question types reflecting realistic scenarios. We further apply a fine-grained evaluation framework of state-of-the-art legal RAG systems, revealing limitations in retrieval, generation, and claim-level analysis in the legal domain.

1 Introduction

The rapid progress of large language models (LLMs) has revolutionized information access, with growing reliance on chatbots shifting semantic search toward a question-answering (QA) paradigm in which users pose questions and LLMs generate responses [1]. This shift can be critical in high-stakes domains such as law, where adoption of LLMs among legal professionals is growing at a rapid pace [4]. Retrieval-augmented generation (RAG) [19] grounds LLM outputs in authoritative sources, increasing response reliability [17]. Despite their benefits, their use in the legal domain remains problematic, even with RAG, due to persistent hallucinations (i.e., generating incorrect information [3]) and a tendency to be perceived as more reliable than they actually are [22, 23].

Theoretical work shows that LLMs must hallucinate to a certain extent, irrespective of their architecture, training data quality, or scale [16]. Empirical evidence reinforces this limitation in the legal domain: Dahl et al. [3] show that general-purpose LLMs hallucinate in approximately 58% to 82% of cases when applied to legal queries. Magesh et al. [22] report hallucination rates of 17% to 33% even for customized legal AI systems, challenging claims that legal research tools are substantially less prone to hallucination. Real-world incidents further emphasize these findings, from early ChatGPT-related failures in legal practice [39] to court sanctions resulting from AI-generated legal filings [6].

The prevalence of hallucinations in both general-purpose and legal-specific LLMs, even with RAG, highlights the need for dedicated legal evaluation benchmarks. Early legal benchmarks such as LEXTREME [28] and LegalBench [11] established important baselines for legal natural language processing (NLP), but were not designed to assess RAG systems, which now dominate the landscape of semantic search and QA.

Legal RAG benchmarks remain limited. LegalBench-RAG [29] focuses on a narrow set of legal retrieval scenarios, while the benchmark of Magesh et al. [22] is specific to U.S. law and relies on costly

manual evaluation. More broadly, existing benchmarks often lack the fine-grained analysis of retrieval and generation, are predominantly English-centric, and primarily target legal professionals rather than supporting the general public in accessing legal information.

To enable fine-grained evaluation of RAG systems, Ferrara et al. [7] introduced the RAG Triad, which assesses context relevance, groundedness, and answer relevance. This framework has since inspired a variety of claim-based evaluation approaches Li et al. [20], Ru et al. [32], where a claim is defined as a minimal, self-contained factual statement [25]. Despite the effectiveness in hallucination detection [33], claim-based approaches remain largely under-explored in the legal domain.

This paper makes three key contributions:

1. We introduce a bilingual benchmark ClaimRAG-LAW ("Fine-grained Claim-level RAG Benchmark for Law") for fine-grained evaluation of legal RAG. The dataset and its generation code are publicly available[1]. Designed to be *diverse*, *representative*, and *fine-grained*, ClaimRAG-LAW covers multiple question categories (adopted from Magesh et al. [22]) and user personas, from legal professionals to lay users seeking access to legal information. It comprises a total of 317 QA pairs sourced from the general data protection regulation (GDPR) [35] (English) and the Luxembourg national civil law [9] (French) (hereafter referred to as *CIVIL*). Additionally, it contains 968 manually-validated claims to support claim-level evaluation of legal RAG systems.
2. We adopt fine-grained metrics proposed by Ru et al. [32] and assess their applicability to legal RAG systems.
3. We benchmark state-of-the-art RAG systems on ClaimRAG-LAW, uncovering domain-specific limitations in retrieval and generation. We further evaluate claim extraction and analysis in legal settings, revealing important limitations, particularly in detecting contradictory claims.

The remainder of the paper is structured as follows. Section 2 surveys related work. Section 3 details the construction of ClaimRAG-LAW. Section 4 reports the results of our empirical evaluation. Section 5 concludes the paper.

2 State of the art

This section reviews two research directions: (1) legal RAG benchmarks, and (2) claim-level evaluation for hallucination detection. We identify two key research gaps that motivate our work: (1) existing legal RAG benchmarks lack comprehensive evaluation compared to general domains; and (2) fine-grained, claim-level evaluation remains largely under-explored in the legal domain despite its success in general-domain RAG.

2.1 RAG Benchmarks in Legal Domain

Early legal benchmarks, including LexGLUE [2], LEXTREME [28], and LegalBench [11] significantly advanced the evaluation of legal NLP and LLM reasoning across diverse tasks and languages. However, these benchmarks primarily focus on classification, prediction, and reasoning tasks rather than RAG. As a result, they are not well suited for assessing retrieval-grounded responses or measuring hallucinations in legal applications.

More recently, dedicated benchmarks for legal RAG evaluation have emerged. CLERC [12] provides a large-scale dataset for case-law retrieval, while LegalBench-RAG [29], which extends LegalBench [11], measures the retrieval precision for contract-related questions. Despite their contributions, both benchmarks are primarily retrieval-centric and do not comprehensively assess end-to-end answer generation.

To address this limitation, LLeQA [21] introduced an expert-annotated French benchmark for long-form statutory QA based on Belgian law, enabling evaluation of retrieve-then-read RAG systems. Nevertheless, its scope remains limited to a single language and jurisdiction. Moreover, it relies largely on holistic automatic metrics and does not provide fine-grained evaluation of generated answers.

[1] Links omitted for double blind.

Several other legal RAG benchmarks have been proposed for specific legal systems, LexRAG [20] evaluates legal consultation dialogues in Chinese law, CBR-RAG [40] focuses on case-based retrieval in Australian law, and KOBLEX [18] targets provision-grounded multi-hop QA in Korean law. While these benchmarks contribute valuable jurisdiction-specific resources, their focus on individual legal systems limits the ability to draw cross-jurisdictional conclusions.

Overall, existing legal RAG benchmarks provide only partial coverage of the challenges involved in retrieval-grounded legal question answering. Although LLeQA partially addresses non-expert information needs, no benchmark systematically evaluates RAG systems from a non-expert perspective across multiple jurisdictions and languages. Furthermore, current datasets offer limited support for fine-grained analysis that jointly examines retrieval quality, answer generation, and hallucination behavior.

In parallel, the broader RAG community has developed evaluation frameworks such as RAGAS [5] and RAGChecker [32], enabling more detailed assessment of retrieval and generation performance. In contrast, equivalent benchmarking resources for legal retrieval-grounded generation remain relatively scarce and are still in an early stage of development.

2.2 Claim-Level Evaluation

The rise of generative AI in law has exposed the limitations of traditional evaluation metrics such as ROUGE and BLEU, which measure lexical overlap but often fail to capture factual correctness and hallucinations in RAG systems [32]. To address this gap, frameworks such as TruLens [7] introduced the RAG Triad, which evaluates context relevance, groundedness, and answer relevance. Although influential in general-domain RAG evaluation, these approaches have seen limited adoption in legal applications.

This limitation is particularly important in law, where answers may closely resemble reference texts while containing subtle factual errors. As noted in LLeQA [21], high semantic similarity does not guarantee factual correctness, making fine-grained, claim-level evaluation essential.

Several frameworks have therefore been proposed to assess factuality at the claim level. FActScore [27] decomposes responses into atomic facts and measures the proportion supported by external evidence. RefChecker [13] extracts subject-predicate-object triples and uses entailment-based reasoning to identify unsupported claims. FENICE [33] combines claim extraction with natural language inference (NLI) to provide interpretable, multi-granular factuality assessments, while SAFE [38] verifies extracted facts against web evidence. More recently, Claimify [26] highlighted the importance of claim extraction itself, showing that inaccurate or incomplete claims can lead to misleading hallucination assessments.

Despite these advances, hallucination detection remains an open challenge [24, 25, 37]. Legal LLMs and RAG systems continue to generate unsupported or misleading statements [3, 22]. An empirical study of RAG-based legal research tools found hallucination rates between 17% and 33%, with *misgrounding* as the dominant failure mode: systems often cite genuine legal sources that do not actually support the generated answer [22]. Such errors are unlikely to be captured by holistic evaluation metrics, underscoring the need for claim-level analysis in legal settings. However, legal-specific methods remain scarce. RePASs (Regulatory Passage Answer Stability Score) [8] introduces an NLI-based metric for evaluating contradiction and obligation coverage in regulatory texts, but its scope is limited to Gulf financial regulations. Consequently, a gap remains for fine-grained, claim-level evaluation of multilingual statutory-law RAG systems.

3 ClaimRAG-Law

We present *ClaimRAG-LAW*, a benchmark for evaluating legal RAG systems. It consists of two complementary datasets: one for end-to-end legal RAG evaluation and another for assessing claim-checking methods.

3.1 Desiderata

The design of ClaimRAG-LAW is guided by three desiderata derived from the gaps identified in the literature.

(1) Diverse. Legal RAG systems serve users with varying levels of expertise and information needs. For example, some users seek general legal guidance, others require precise factual references, while some may pose questions based on incorrect assumptions. To capture this diversity, ClaimRAG-LAW includes QA pairs spanning four question categories inspired by Magesh et al. [22]:

(1) *General legal research questions* represent a typical use case for legal RAG systems, where users ask general questions about, e.g., common-law doctrines, specific holdings of court cases, or interpretations of regulations.

(2) *Factual recall questions* target verifiable details that require minimal legal interpretation, such as citations, effective dates of legislation, or specific entities mentioned in a clause.

(3) *False premise questions* begin from an incorrect assumption about a legal fact and therefore require the system to identify and correct the erroneous premise.

(4) *Jurisdiction or time-specific questions* capture the challenge that laws vary across jurisdictions and evolve over time.

These questions are designed to represent the needs of legal experts, public officials with partial legal knowledge, and laypersons with limited ability to verify generated answers.

(2) Representative. Evaluating whether RAG systems generate legally grounded answers, as opposed to relying on parametric knowledge, requires coverage across multiple jurisdictions and languages. To this end, ClaimRAG-LAW draws its QA pairs from two complementary legal sources: (1) the General Data Protection Regulation (GDPR) [35], a supranational European regulatory framework (English version), and (2) a national Civil Code (CIVIL) [9] (French version). These sources provide variation in both jurisdictional scope and language, enabling a more representative evaluation of legal RAG systems.

(3) Fine-grained. Recent RAG evaluation research increasingly relies on claim-level analysis to assess factual grounding and hallucinations. However, most approaches assume that claim extraction and verification are sufficiently reliable. To support fine-grained legal RAG evaluation, ClaimRAG-LAW includes a claim-level dataset derived from approximately half of the QA pairs. Claims were automatically extracted and subsequently validated by a legal expert, who also annotated their logical relation to the supporting legal context. This enables detailed analysis of retrieval quality, factual grounding, and hallucination behavior in generated legal answers.

3.2 The dataset

Following the design principles outlined above, we constructed ClaimRAG-LAW using a two-stage methodology. First, QA pairs were generated automatically from the source regulations. Second, all generated QA pairs were manually reviewed and validated by a legal expert. Additional details on the generation and validation procedures are provided in Appendix **??**.

ClaimRAG-LAW comprises 317 legal QA pairs drawn from two complementary legal sources: GDPR (English) and CIVIL (French). Table 1 summarizes the dataset statistics. Fig. 1 shows example questions from different categories together with their corresponding ground-truth answers from ClaimRAG-LAW (GDPR). More examples are provided in Section **??**.

To support fine-grained evaluation, we additionally constructed a claim-level dataset from a subset of the QA pairs. In total, ClaimRAG-LAW contains 968 manually validated claims, of which 86.6% (838/968) were judged by the legal expert to be correct with respect to the source legal context. This claim-level annotation enables detailed analysis of factual grounding and hallucinations in legal RAG systems.

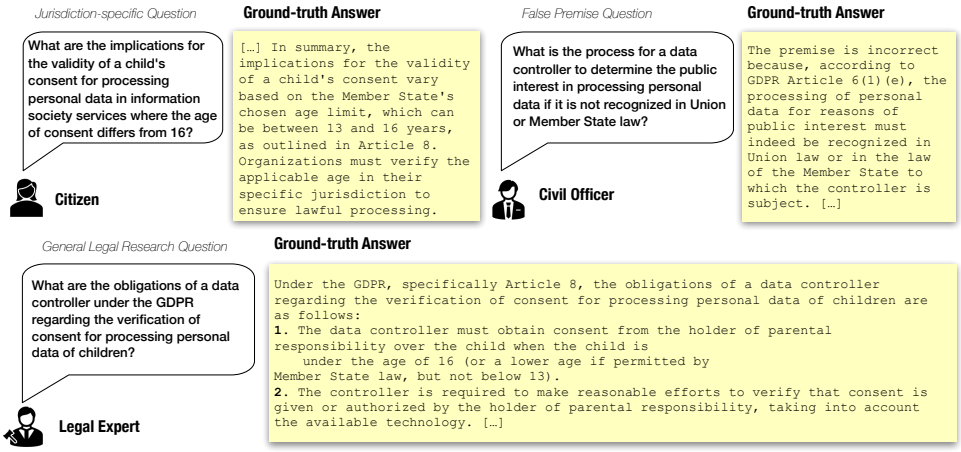


Figure 1: Example of GDPR questions and ground-truth answers in ClaimRAG-LAW.

Table 1: Statistics for ClaimRAG-LAW Dataset

		Persona			**Question Category**				**Claims**					
	$\sum_{QA}$	P_1	P_2	P_3	CG_1	CG_2	CG_3	CG_4	$\sum_{A}$	$\sum_{C}$	Valid	ϵ	η	ζ
GDPR (EN)	149	20	77	52	131	9	8	1	63	520	453	295	151	7
CIVIL (FR)	168	35	29	104	137	24	5	2	89	448	385	284	77	24
Total	**317**	**55**	**106**	**156**	**268**	**33**	**13**	**3**	**152**	**968**	**838**	**579**	**228**	**31**

[1] *$\sum_{QA}$ refers to the number of QA pairs. P_1, P_2, P_3 refer to the personas: citizen, civil officer, and legal expert, respectively. CG_1, CG_2, CG_3, CG_4 refer to the question categories: general legal research, factual recall, false premise, jurisdiction/time-specific, respectively.*

[2] *$\sum_{A}$: the number of generated answers used to generate the claims, and $\sum_{C}$: the number of claims. Valid indicates the number of claims deemed correct by the expert, and ϵ, η, ζ refer to entailment, neutral, and contradiction, respectively.*

4 Benchmarking RAG Systems in Legal Domain

4.1 Research Questions

RQ1. How well do existing RAG systems perform in legal settings? This RQ provides a fine-grained assessment of RAG systems in legal settings by assessing both their overall performance on our curated dataset and the performance of their retrieval and generator components separately. To achieve this, we use *RAGChecker* [32], a fine-grained RAG evaluation framework. We selected RAGChecker for two reasons. First, it has shown to provide a detailed view of the performance of RAG systems. Second, Ru et al. [32] report that its metrics correlate more strongly with human judgments in terms of correctness (both Pearson and Spearman) compared to alternative frameworks such as RAGAs [5]. This alignment with human evaluation is particularly important in the legal domain, where subtle differences may not be captured by aggregated metrics.

RQ2. How accurate are automated claim extraction and verification methods in the legal domain? Since claims form the basis of fine-grained RAG evaluation, this RQ assesses the reliability of claim-level analysis in the legal domain. Specifically, we evaluate the accuracy of RefChecker [13] in extracting legal claims from the generated responses and verifying their entailment relationship with a reference text. We focus on RefChecker because it serves as the claim-level analysis engine underlying RAGChecker, which is used in RQ1.

4.2 Accuracy of RAG Systems in Legal Settings (RQ1)

To answer RQ1, we evaluated the performance of several RAG systems on ClaimRAG-LAW using the fine-grained evaluation framework proposed by Ru et al. [32].

RAG Systems. We evaluated eight RAG systems obtained by combining two retrieval methods with four LLM-based generators. For retrieval, we used the same retrievers considered by Ru et al. [32]: *BM25* [31], a standard sparse retriever, and *E5-Mistral-7B-Instruct* [36], a state-of-the-art dense retriever. For generation, we adopted three of the models used by Ru et al. [32]: *GPT-4o* [14], *Llama-3.1-8B-Instruct* [10], and *Mixtral-8x7B* [15]. We excluded *Llama-3.1-70B-Instruct* because of its high computational cost. In addition, we included GPT5 [34] to assess whether a newer generation of the same model family provides advantages in legal QA.

Implementation Details. Following Ru et al. [32], we implemented both retrievers using OpenSearch index (`https://opensearch.org/`) and E5-Mistral-7B-Instruct tokenizer for chunking. We split documents into fixed-size chunks of 600 tokens with an overlap of 120 tokens. We prefer fixed-size chunking over article-level chunking because legal documents, such as GDPR and CIVIL, exhibit substantial variation in article length, resulting in inconsistent retrieval granularity. We chose 600-token chunks based on the ablation study of Ru et al. [32], which showed consistent improvements in claim recall and F1 as chunk size increases.

We retrieved the top-$k = 10$, corresponding to 6000 tokens of retrieved context. Together with the system prompt, question, and generated response, this remained within the 8192-token context window of Llama-3.1-8B-Instruct, the most constrained generator in our study. For generation, we used the default prompt template of Ru et al. [32], a temperature of 0.0, and a single evaluation run. We conducted all experiments on a workstation with a single NVIDIA RTX 3090 GPU (24 GB), an AMD Ryzen 9 5900X 12-core CPU, and 64 GB RAM.

RAGChecker Input Construction. RAGChecker requires a tuple consisting of a question, retrieved chunk (context), generated response, and ground-truth answer. In ClaimRAG-LAW, the question and ground-truth answer are provided, whereas the retrieved chunk and generated response are produced by each RAG system. For every question, the system retrieves the top-k relevant chunks from the corresponding legal document (GDPR or CIVIL) and generates a response. The resulting tuples are then evaluated with RAGChecker.

Metrics. We use the fine-grained metrics proposed by Ru et al. [32]. These metrics are based on claim extraction and entailment analysis between claims and reference texts.

Overall metrics include (i) *precision (P)*, the proportion of response claims entailed by the ground-truth answer; (ii) *recall (R)*, the proportion of ground-truth claims correctly identified by the model response; and *F1 score*, the harmonic mean between P and R.

Retriever metrics include (i) *claim recall (CR)*, the proportion of ground-truth claims entailed by the retrieved chunks; and (ii) *context precision (CP)*, the proportion of retrieved chunks entailing at least one ground-truth claim.

Generator metrics include (i) *faithfulness (FT)*, the proportion of response claims entailed by the retrieved chunks; (ii) *self knowledge (SK)*, the proportion of correct response claims not entailed by the retrieved context; (iii) *context utilization (CU)*, the ratio of correct response claims supported by the retrieved chunks to the total relevant ground-truth claims entailed by the retrieved chunks; (iv) *hallucination (HL)*, the proportion of incorrect response claims not entailed by the ground-truth answer or any of the retrieved chunks; (v) *relevant noise sensitivity (NS_{r})*, the proportion of incorrect response claims entailed by relevant chunks; and (vi) *irrelevant noise sensitivity (NS_{ir})*, the proportion of incorrect response claims entailed by irrelevant chunks. A chunk is deemed *relevant* if it entails any claim in the ground-truth answer, and *irrelevant otherwise*.

Results. Table 2 reports the average results across all question-answer pairs. For comparison, we also include the best-performing RAG system by Ru et al. [32], evaluated on several non-legal datasets.

Overall, most RAG systems achieved higher P, R, and F1 scores on the FR subset than on the EN one. Since legal applications prioritize correctness over completeness, precision is particularly important: providing incorrect legal information can have more severe consequences than omitting relevant details,

Table 2: Performance of baseline RAG systems evaluated using *RAGChecker* on two legal datasets. Arrows indicate preferred direction (↑ higher is better, ↓ lower is better).

		Overall			Retriever		Generator					
RAG System	Lang†	**P↑**	**R↑**	**F1↑**	**CR↑**	**CP↑**	**CU↑**	**NS_r↓**	**NS_{ir}↓**	**HL↓**	**SK↓**	**FT↑**
BM25+Llama3-8B	EN	45.0	62.1	48.2	82.9	**76.9**	69.0	48.5	2.0	4.6	2.1	**93.3**
BM25+Llama3-8B	FR	43.6	45.5	38.3	65.1	28.2	46.1	24.8	13.7	17.9	5.9	76.1
BM25+Mixtral-8x7B	EN	49.1	68.6	52.2	82.7	75.4	75.9	45.3	2.2	3.4	3.4	93.1
BM25+Mixtral-8x7B	FR	48.2	52.9	44.8	64.7	28.3	54.1	20.9	14.5	16.4	4.2	79.5
BM25+GPT-4	EN	54.3	77.7	**60.9**	82.2	76.8	83.1	40.6	2.3	2.9	4.7	92.4
BM25+GPT-4	FR	65.6	82.7	**70.3**	90.5	**34.6**	85.4	27.2	**2.5**	4.1	1.5	93.9
BM25+GPT-5	EN	49.6	**78.2**	56.8	**86.1**	70.3	**86.6**	47.5	**1.4**	**1.5**	**1.6**	90.9
BM25+GPT-5	FR	66.4	**84.2**	69.9	**90.8**	31.8	**86.9**	28.7	3.7	**1.2**	**0.7**	**98.1**
E5+Llama3-8B	EN	47.8	61.4	50.0	65.7	61.9	71.6	31.1	11.6	9.5	6.4	84.1
E5+Llama3-8B	FR	52.6	64.5	52.8	80.0	29.7	70.2	23.7	12.5	11.2	3.5	85.4
E5+Mixtral-8x7B	EN	49.0	56.0	47.9	66.6	63.1	72.5	36.1	10.2	4.7	3.4	91.9
E5+Mixtral-8x7B	FR	64.5	66.0	61.4	80.6	29.0	73.9	21.1	9.7	4.7	1.1	94.2
E5+GPT-4	EN	**59.8**	63.2	58.3	65.4	61.7	74.5	28.8	8.4	3.0	7.9	89.1
E5+GPT-4	FR	**69.7**	66.6	64.9	80.3	28.5	74.7	**18.6**	8.5	3.3	2.7	94.0
E5+GPT-5	EN	59.5	69.0	60.7	65.4	53.7	80.6	**26.2**	8.6	5.6	8.6	85.7
E5+GPT-5	FR	64.5	72.5	65.0	79.3	23.8	77.8	20.2	10.3	4.9	3.7	91.4
Ru et al. [32]‡		62.0	53.0	52.7	83.5	61.8	60.4	28.9	3.5	5.7	1.4	92.9

† EN = GDPR and FR = CIVIL

‡ *best reported results by Ru et al. [32] on generic domain using E5+GPT-4, obtained using datasets covering only English language queries and corpora.*

which users may still recover through additional search.

Under this criterion, E5+GPT4 achieved the best overall performance, marginally outperforming E5+GPT5; this confirms the results in Ru et al. [32]. Although BM25+GPT5 yielded the highest recall, its lower precision, especially on EN questions, makes it less attractive for legal use. We also remark that all systems achieved substantially higher scores on ClaimRAG-LAW than those reported by Ru et al. [32]. A likely explanation is the difference in dataset scale: ClaimRAG-LAW contains 394 questions derived from two legal documents, whereas the benchmark of Ru et al. [32] comprises 4162 questions spanning more than one million documents.

The retrieval results reveal that BM25 consistently outperforms E5 in both CR and CP across languages. This suggests that *dense retrievers may excel in retrieving semantically related contexts, but do not necessarily identify legally relevant ones*. In contrast, BM25's lexical matching appears better aligned with legal terminology and document structure. This observation explains that keyword-based retrieval methods remain highly competitive in legal applications [30].

We also observed relatively low CP values for CIVIL despite the high CR value achieved on both datasets. A likely reason is that CIVIL articles are typically shorter than GDPR articles, causing individual chunks to span multiple articles and thus contain both relevant and irrelevant content. This result highlights that *retrieval effectiveness depends not only on the number of retrieved chunks (k), but also on their legal relevance and granularity.*

The effectiveness of the generation component is reflected in the faithfulness (FT) and context utilization (CU) metrics. GPT-4o and GPT-5 paired with BM25 achieved particularly strong results, with FT exceeding 90% and CU exceeding 80% in both languages. Other models exhibited greater variability. For example, Mixtral-8x7B achieved substantially higher FT in FR when paired with E5 (94.2%) than with BM25 (79.5%), while it seemed less affected for EN. Similarly, Llama-3.1-8B benefitted from E5 in FR (from 76.1% to 85.4%) but performed worse in EN (from 93.3% to 84.1%). CU results generally improved for these two models paired with E5, except for a small drop for Mistral-8x7b for EN.

Table 3: Claim verification against expert labels. P: Precision (%); R.: Recall (%); F1: F1-score (%).

	GDPR (EN)				CIVIL (FR)			
	$\sum_C$	P	R	F1	$\sum_C$	P	R	F1
ϵ	295	84.6	73.4	78.6	284	89.3	82.4	85.7
η	151	68.1	72.2	70.1	77	53.3	62.3	57.5
ζ	7	2.6	14.3	4.4	24	24.2	33.3	28.1

$\sum_C$ is the number of claims per class in ClaimRAG-LAW according to the expert's manual analysis; ϵ, η, ζ refer to entailment, neutral, and contradiction, respectively.

SK and HL results provide a complementary view of the observations above. GPT-5 achieved the highest SK scores while also maintaining the lowest HL rates. In general, systems that rely more heavily on retrieved evidence (high CU) tend to hallucinate less. Conversely, when generators depend more on internal knowledge, hallucinations become more frequent. This pattern is particularly evident for BM25+Llama-3.1-8B and BM25+Mixtral-8x7B, whose higher HL rates (17.9 and 16.4, respectively) coincide with lower CU values, especially on FR questions. The markedly higher HL rates in FR are also consistent with lower CP and FT scores, suggesting that weaker retrieval quality increases the likelihood of hallucinations.

Finally, NS results show that incorrect claims arise predominantly from partially relevant chunks rather than entirely irrelevant ones. Across all systems, NS_r was consistently higher than NS_{ir}, indicating that generators are more likely to misinterpret or over-generalize information from relevant contexts than to fabricate content from unrelated passages.

4.3 Accuracy of Claim Analysis in Legal Settings (RQ2)

To assess the reliability of claim analysis in the legal domain, we evaluated RefChecker [13] on ClaimRAG-LAW. Consistent with its original implementation, we used GPT-4-0613 as the enabler LLM.

Our evaluation comprises two steps: (i) assessing the accuracy of claim extraction from generated responses and (ii) assessing the accuracy of claim verification, namely determining the entailment relationship between a claim and a reference text.

Evaluation Procedure. During the construction of ClaimRAG-LAW, we used RefChecker to extract the claims from 152 answers, producing 968 claims in total. These claims were manually validated by the legal expert (see Appendix **??**), who labeled each claim as valid or invalid.

For claim extraction, we computed *Extraction Accuracy* as the proportion of valid claims among all extracted claims. Although this metric primarily reflects precision, rather than the ability to identify missed claims, it remains informative for assessing the reliability of claim analysis as an intermediate step in fine-grained RAG evaluation.

For claim verification, RefChecker assigns one of three labels to each claim: *entailment* indicating whether the claim extracted from the answer is entailed in the retrieved contexts, *contradiction* when the claim contradicts the contexts, and *neutral* otherwise. To isolate the verification component, we bypassed claim extraction and provided RefChecker with the expert-validated claims (453 for GDPR and 385 for CIVIL) together with the retrieved contexts.

Results. For claims extraction, RefChecker achieved an accuracy of 98% (452/461) for GDPR (EN) and 98.7% (385/390) for CIVIL (FR). These results indicate that *RefChecker can reliably decompose legal responses into valid claims across both EN and FR.*

Table 3 reports the performance of RefChecker's verification labels against the expert annotations. The results reveal a clear weakness in identifying contradictory claims. The average F1 score for the *contradiction* class is only 4.4% on GDPR and 28.1% on CIVIL. The corresponding precision and recall values, particularly on GDPR, indicate that RefChecker both misses many genuine contradictions and incorrectly labels non-contradictory claims as contradictions. As a result, errors in claim verification may propagate to the downstream evaluation of RAG systems.

Although contradictory claims are relatively rare in our dataset, the results suggest that contradiction

detection remains a challenging problem in legal settings. *Overall, while RefChecker performs very well at claim extraction, its verification component, especially for contradiction detection, requires further improvement for reliable legal-domain evaluation.*

4.4 Threats to Validity and Limitations

Threats to Validity. Some question categories like false-premise are underrepresented (5.4% in GDPR; 3.0% in CIVIL), limiting the strength of conclusions for these cases. In addition, the dataset includes only two languages and one document per language, limiting linguistic and document-level diversity. All QA pairs were generated using GPT-4, which is also one of the evaluated generators and may therefore benefit from generation bias.

The manual validation was performed by a single legal expert. Although annotation quality was supported through a pilot phase, detailed guidelines, and iterative feedback, the absence of multiple annotators prevents the computation of inter-annotator agreement metrics.

Limitations. As a synthetic dataset, ClaimRAG-LAW may not fully capture the diversity and complexity of real-world legal queries, which may limit the generalizability of our findings. Moreover, although the dataset is suitable for benchmarking legal RAG systems, it should not be used for legal advice without expert review.

5 Conclusion

This paper has introduced ClaimRAG-LAW, a bilingual, bi-jurisdictional benchmark for fine-grained legal RAG evaluation. The benchmark contains 317 expert-validated QA pairs, spanning diverse question categories and user personas, together with 968 manually validated claims. Using ClaimRAG-LAW, we have benchmarked eight state-of-the-art RAG systems and identified domain-specific weaknesses in both retrieval and generation that are not captured by existing legal RAG benchmarks. We also showed that current claim-level evaluation frameworks, while effective in general-domain settings, exhibit important limitations in legal contexts, particularly for contradiction detection. Future work includes a deeper investigation of hallucinations in legal RAG systems, expanding underrepresented question categories, and enriching ClaimRAG-LAW with additional expert-authored QA pairs and claims.

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